

Aggregate analysis

1. multipop stack
2. incrementing binary counter.

1 0 1 1 0 1 1 1



1 0 1 1 1 0 0 0

A is representation, k - bit counter.

Increment (A, k)

$\bar{i} = 0$

while $\bar{i} < k$ and $A[\bar{i}] == 1$

└ $A[\bar{i}] = 0$
 $\bar{i} = \bar{i} + 1$

if $\bar{i} < k$

$A[\bar{i}] = 1$



1 0 1 1 0 1 1 1

Running time: $O(k)$

sequence of n operations, ^{increments / starting from 0.} running time

$O(kn)$

can we get better??

A	7	6	5	4	3	2	1	0
0	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	1
2	0	0	0	0	0	0	1	0
3	0	0	0	0	0	0	1	1
4			0	0	0	1	0	0
5				0	0	1	0	1
.		0		0	0	1	1	0
.				0	0	1	1	1
.				0	1	0	0	0
n								

$$A[0] = n$$

$$A[1] = \frac{n}{2}$$

$$A[2] = \frac{n}{2^2}$$

$$A[3] = \frac{n}{2^3}$$

$$A[k-1] = \frac{n}{2^{k-1}}$$

$$+ O(n)$$

amortized cost per operation

$$\frac{O(n)}{n} = O(1)$$

Dynamic array

Advantage of an array

- access each element takes $O(1)$ time.
- Simple.

Disadvantage

- fixed size.
- Dynamic operations are costlier

Data
size
capacity

→

2	5	3	1	6	9						
---	---	---	---	---	---	--	--	--	--	--	--

6
10

size : $\leftarrow$ # elements currently stored
in the array

capacity : $\leftarrow$ total # elements the array
can accommodate.

Insertion in dynamic array

2	5	1		
---	---	---	--	--

insert 6

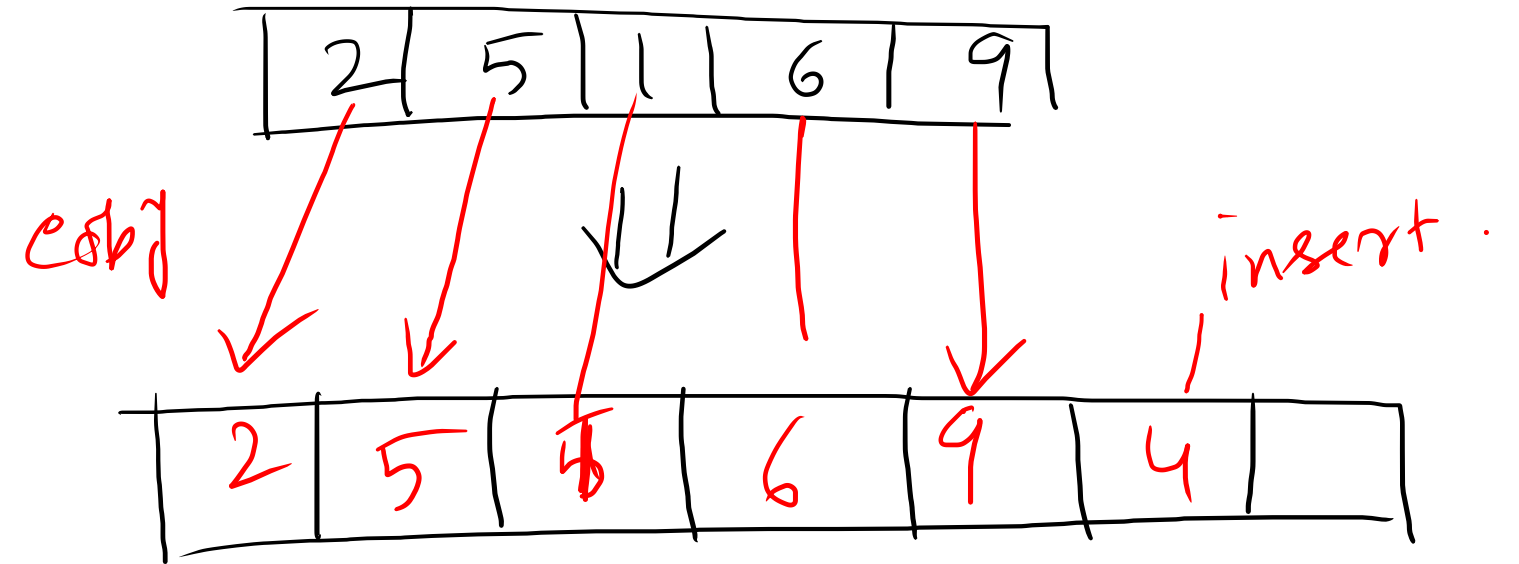
2	5	1	6	
---	---	---	---	--

insert 9

2	5	1	6	9
---	---	---	---	---

insert 4

size == capacity



create a new array
of capacity > old capacity.

Q: How much increment?

new capacity $\neq$ old capacity + 1

Dynamic array insert (A, m, n, data)

$m \leftarrow \text{capacity}$
 $n \leftarrow \text{size}$

if $m == n$

create a new array newA of capacity $m + 1$

for $i = 1$ to n

└─ newA[i] = A[i]

$m = m + 1$

└─ replace A by newA

$n = n + 1$

A[n] = data

$O(n)$

Amortized cost per operation:

	capacity	size	cost
	0	0	0
insert(1)	1	1	c
insert(2)	2	2	c + c
insert(3)	3	3	2c + c
...			
...			
(i)	i	i	(i-1)c + c
...			

amortised cost per operation:
 $\frac{O(n^2)}{n} = O(n)$

$$c \sum_{i=1}^n 1 + c \sum_{i=1}^n 1+2+3+\dots+n-1 = O(n^2)$$

What about $\text{new capacity} = \text{old capacity} + K$

K : constant.

capacity	size	cost
0	0	0
K	1	c
	$\vdots$	$\vdots$
K	K	c
$2K$	$K+1$	$cK + c$
	$\vdots$	$\vdots$
$2K$	$2K$	c
$3K$	$2K+1$	$2cK + c$

$$\sum_{l=0}^n c + \sum_{l=1}^{n/k} l c k$$

$$= cn + ck \left[1 + 2 + \dots + \frac{n}{k} \right]$$

$$= cn + ck \cdot \frac{\frac{n}{k} \cdot \left(\frac{n}{k} + 1 \right)}{2}$$

$$= cn + O(n^2)$$

$$= O(n^2)$$

amortized cost per operation is $\frac{O(n^2)}{n} = O(n)$

